

Lab 8: Cluster Analysis

Evaluation: 5% of the total grade

In this lab you will perform clustering on three datasets provided each having two features. As the first step of data understanding, visually examine the scatter plots of all three datasets for the apparent number of clusters in each dataset. Then, perform the clustering tasks on these datasets using two algorithms as outlined below:

For dataset1:

1. perform k-means clustering on the dataset.
2. determine the optimal number of clusters based on silhouette analysis using k-means clustering algorithm.
3. compute and display clustering results for different number of clusters (k) along with corresponding silhouette graphs and scores.
4. compute and display dendrogram for the agglomerative hierarchical clustering algorithm cut at the level of optimal k clusters determined in step 2.

For all three datasets dataset1—dataset3:

1. perform agglomerative hierarchical clustering using four types of linkages, i.e., single, averaging, complete, and ward linkage.
2. display subplots comparing clustering results in a grid for 4 linkages, 3 datasets (using appropriate number of clusters for each of the datasets).

Briefly explain your results / observations.

Submit your lab report along with the source code preferably as Jupyter notebook in pdf format by the due date as per schedule on the Nexus.

Lab Grading Criteria

Fraction	0.8 – 1	0.4 – 0.7	0 – 0.3	Max Score
Category				
Model development workflow	Followed all steps in the modeling process	Followed most steps in the modeling process	No order in the modeling process	1
Tasks and observations	Performed all the tasks correctly, results properly presented and well explained	Performed most of the tasks correctly, results fairly presented and explained	Performed some of the tasks correctly, poorly presented results with little explanation	3
Code	Well organized, easy to follow and clean code with meaningful comments	Somewhat organized, a little hard to follow and fairly clean code with some comments	Not organized, very hard to follow and messy code with little or no comments	1
Total				5